

How to solve?

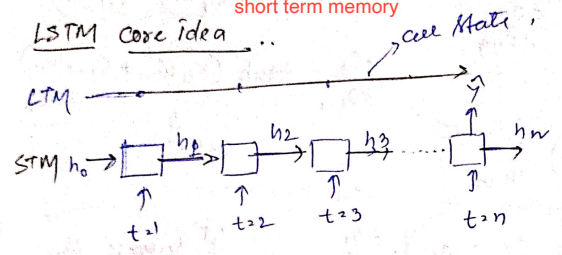
- 1) Gradient clipping.
- 2) Controlled LR.
- 3) LSTM.

7/08/25

LSTM

fullform- long-short term memory
important for later advanced topics like LLM
Long term memory.
short term memory

LSTM Core Idea ..

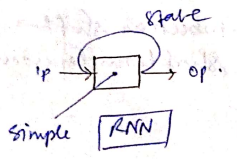


(RNN)

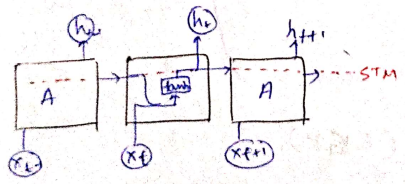
designed to handle long sequences of data effectively

RNN only has STC

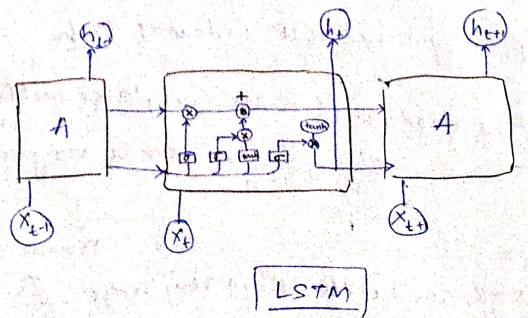
Two separate memory lines.
-long term context LTC
-short term context STC



cell state->LTM
hidden state->STM

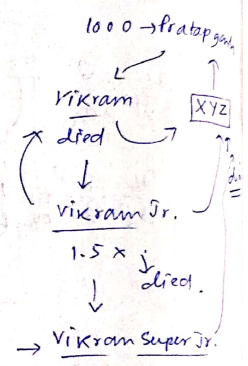


RNN



LSTM

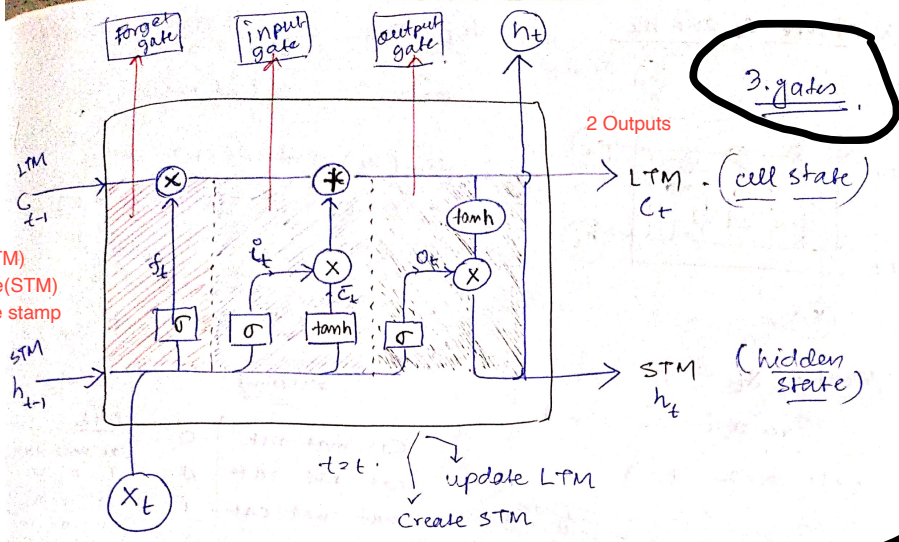
story example



Complex architecture for communication between LTM and STM

At each time step,

- 3 Inputs,
-previous cell state(LTM)
-previous hidden state(STM)
- input for current time stamp

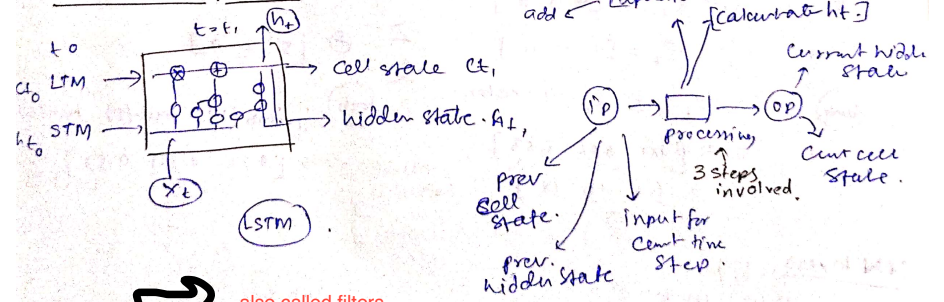


LSTM gates

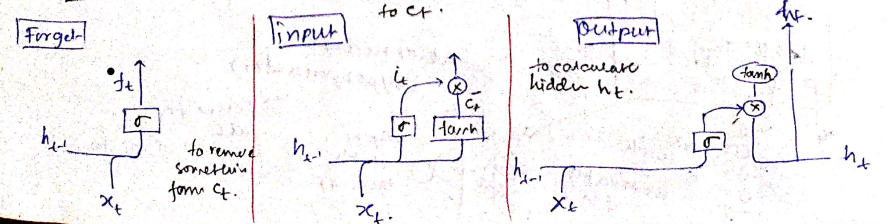
Very important

- forget** => decides what things to delete from LTM.
- input** => decides what new things should add in LTM.
- output** => decides what to give as output based on the LTM. (Y/N).

The Architecture

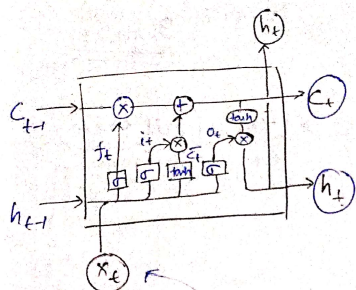


The Gates



All different Inputs are concatenated before processing

What are c_t and h_t

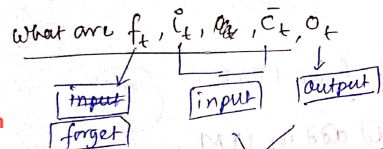


What is x_t input word

vector $[\dots]$

$t=1 \quad t=2 \quad t=3$

4 other vectors



found in

vectors

c_t, h_t dim same
 f_t, i_t, o_t

pointwise operations

(X)

$c_{t-1} = [4 \ 5 \ 6]$

(+)

$f_t = [1 \ 2 \ 3]$

(tanh)

$c_{t-1} \otimes f_t \rightarrow \text{vector}$

point wise multiplication

$= [4 \times 1 \ 5 \times 2 \ 6 \times 3]$
 $= [4 \ 10 \ 18]$

point wise addition, pointwise tanh opern

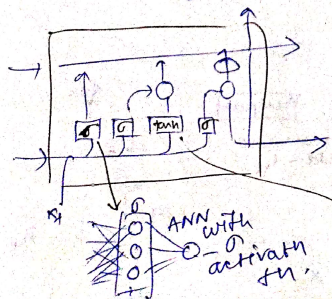
$= [5 \ 7 \ 9]$

(tanh)

$[\tanh(4) \ \tanh(10) \ \tanh(18)]$
 $= [0.26 \ 0.34 \ 0.53]$

NN layers

$\sigma \ \sigma \ \tanh \ \sigma$



no. of nodes (hyperparameter)

tanh

ANN with tanh AF

same for all $\rightarrow \sigma, \sigma, \tanh, \sigma$

$h_t \ c_t$
vectors
 $[0.1 \ 0.3 \ 0.9]$
 $\{3d \ vector\}$
 $\dim(h_t) = \dim(c_t) \ w$
 $[0.1 \ 0.95 \ 0.6] \ [0.55 \ 0.6 \ 0.0]$
 $h_t \ c_t$

Example

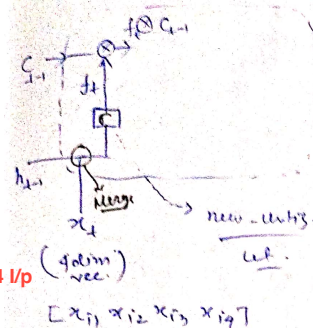
Sentence

Cat	mat	rat	0
cat	rat	rat	0
mat	mat	cat	1

$[0 \ 0] \ [0 \ 1 \ 0] \ [0 \ 0 \ 1]$

Word vector

The forget gate



Assume 4 1p vectors

$[x_{i1} \ x_{i2} \ x_{i3} \ x_{i4}]$

$(3 \times 7) \ (7 \times 1)$
 $f_t \rightarrow (3 \times 1)$

$$f_t = \sigma(w_f [h_{t-1}, x_t] + b_f)$$

Intuition how forget gate does removal/forget

$f_t \otimes c_{t-1}$

$(c_{t-1}) \rightarrow \text{remove}$

$\rightarrow [2 \ 2.5 \ 3]$
 $\{ \text{values reduced} \}$
 $\{ \text{info. lost} \}$
 $\{ 50\% \text{ info reduced} \}$

input output $\rightarrow$ cell state remove

(0) (0) (0)

Forget gate works in 2 stages

1) calculate f_t
2) $c_{t-1} \otimes f_t$
 $f_t = [f_1 \ f_2 \ f_3]$
 $\dim = 3$

$$f_t = \sigma(w_f [h_{t-1}, x_t] + b_f)$$

merge concatenation of $h_{t-1} \ x_t$

$$f_t = \sigma(w_f [h_{t-1}, x_t] + b_f)$$

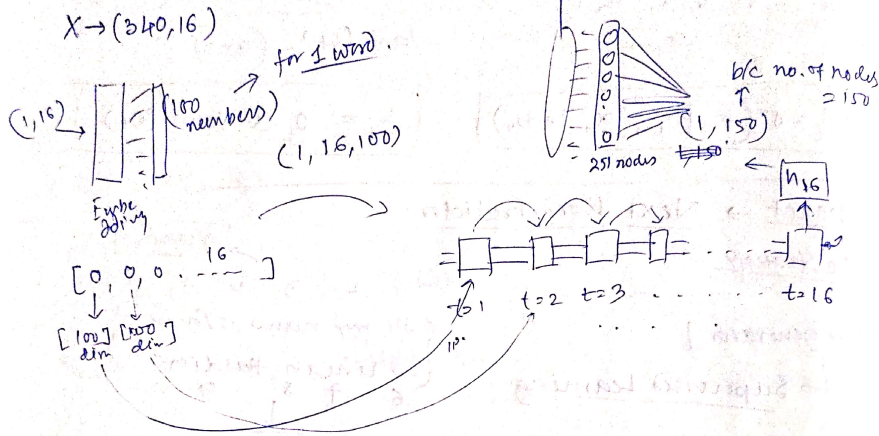
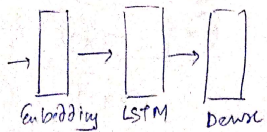
The input gate

add some new imp. info to cell state
Input gate works in 2 stages

Stages

1) $\bar{c}_t$ candidate cell state
2) i_t calculate
3) calculate c_t

The Architecture



How to improve performance?

- 1) more data
- 2) hyperparameter tuning
- 3) Advanced architectures
 - Stacked LSTM
 - Bidirectional LSTM
 - Transformers

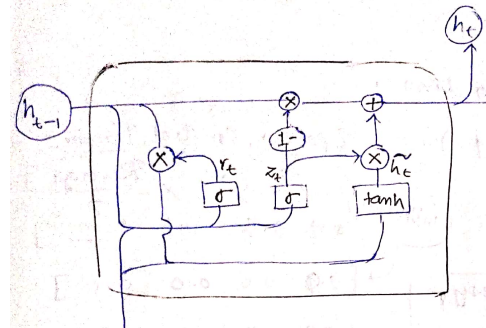
MultiLayer LSTM

Big idea behind GRU. → no all state. only 1 → 'hidden state'.

mother carry long term & short term context

→ NO. 3 gates → only 2 gates here.

[reset. update.]



space

h_{t-1} → h_t

IP. OP.

These are all vectors

vectors.

candidate hidden state

reset gate

update gate

- decides what to forget from past memory

update gate

- decides how much update to keep from the past

update gate

- decides how much update to keep from the past

Except input x_t all same dimension

dim same.

[NN layers]

Same on no. of nodes.

prev. hidden state
curr. input

$[h_{t-1}, h_t, x_t, r_t, z_t, \tilde{h}_t]$

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

candidate hidden state

reset gate

update gate

Architecture

Input Output

$h_{t-1}, x_t \rightarrow h_t$

1) Calculate r_t (reset gate)

2) Calculate $\tilde{h}_t$ (candidate hidden state)

3) Calculate z_t (update gate)

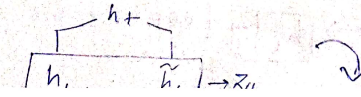
4) Calculate h_t (current hidden state)

made recently in 2014 simpler and faster

to train and inference than LSTM.

has less no of parameters compared to LSTM.

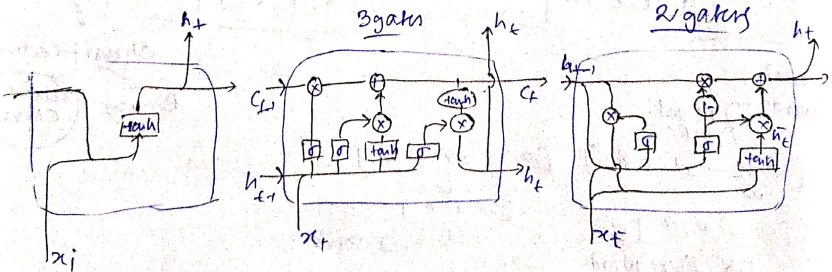
but performance may slightly be less



9/08/25

GRU - Gated Recurrent Unit

a type of RNN



Simple RNN

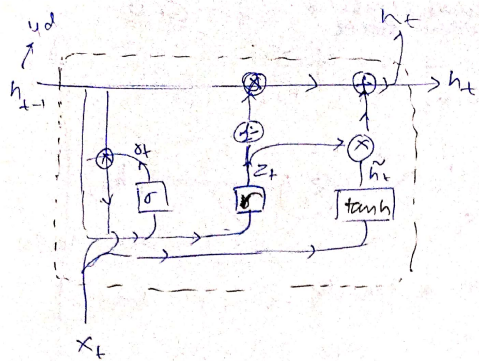
LSTM

GRU

97

2014

less no. of param



$$z_t = \sigma(w_z [h_{t-1}, x_t] + b_z)$$

$$\tilde{h}_t = [0.7 \quad 0.2 \quad 0.1 \quad 0.2]$$

$$h_{t-1} = [0.6 \quad 0.6 \quad 0.7 \quad 0.1]$$

$$\text{let } z_t = [0.1 \quad 0.7 \quad 0.8 \quad 0.2]$$

$$h_t = ((1 - z_t) \otimes h_{t-1}) \oplus (z_t \otimes \tilde{h}_t)$$

if $z_t \uparrow \Rightarrow \tilde{h}_t$ more value
 $z_t \downarrow \Rightarrow h_{t-1}$ more value.

Suppose $z_t = 1$

$$h_{t-1} = [0.6 \quad 0.6 \quad 0.7 \quad 0.1]$$

$$(1 - z_t) = [0.9 \quad 0.3 \quad 0.2 \quad 0.8]$$

$$\therefore (1 - z_t) \otimes h_{t-1} = [0.54 \quad 0.18 \quad 0.14 \quad 0.08]$$

$$z_t \otimes \tilde{h}_t = [0.07 \quad 0.14 \quad 0.08 \quad 0.04]$$

$$\therefore h_t = [0.61 \quad 0.32 \quad 0.22 \quad 0.12]$$

power. conflict largely average

Diff. b/w LSTM

parameter count

for input size d & hidden size h ,
 no. of parameters =
 $4 \times (d \times h) + (h \times h) + n \times 4 \times (d \times h) + (h \times h) + h$

& GRU:

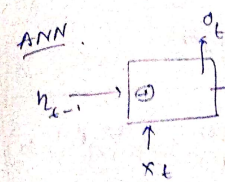
for same size -

$$3 \times ((d \times h) + (h \times h) + h) \times 3$$

$$+ ((d \times h) + (h \times h) + h)$$

What is Deep RNN?

ANN

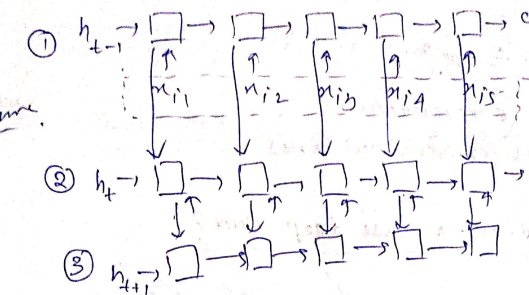


Sentiment

review

5 words +ve -ve

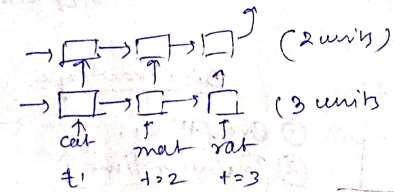
unfolding in time.



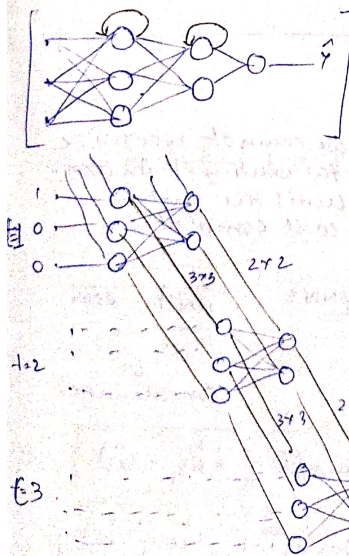
Architecture

stacked vertically,
 O/p from first RNN
 inputs to Next RNN

Architecture



Breaking it down

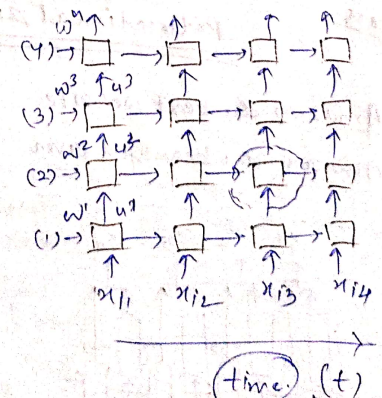


review	sentiment
cat. mat. rat	1
rat. rat. mat.	0
mat. cat. cat.	1

[1 0 0] [0 1 0] [0 0 1]
 [t=3] 3 words

Notation

For studying properly in future

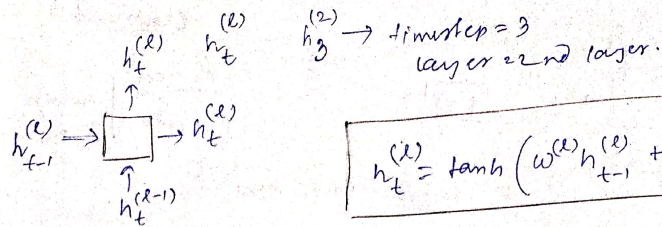


depth
 (l)

vertically,
 also represents
 the no of layers

horizontally represents
 sequence time steps.

time (t)

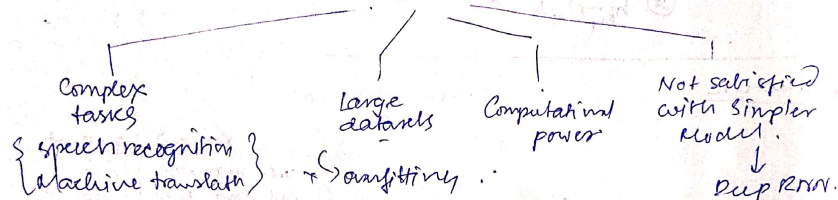


$$h_t^{(l)} = \tanh(w^{(l)} h_{t-1}^{(l)} + u^{(l)} h_t^{(l-1)} + b)$$

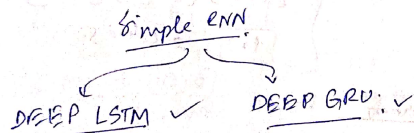
Why and when to use?

1. Hierarchical Representation . for multiple level of abstraction
2. Customization for advanced tasks.

When to use deep RNN?



Variants



Disadvantages

- ① Overfitting.
- ② Training time ↑

code
return_sequences=True
in hidden RNN
and False for
last layer RNN

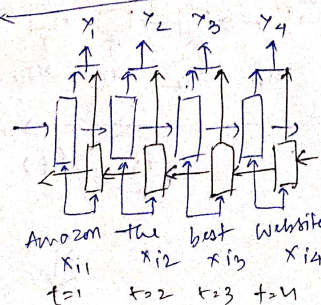
11/08/2025

Bidirectional RNN

NER
Entity
Recognition

Amazon the best website.
Amazon the beautiful river.

→ we cannot recognize the identity of Amazon until the sentence will be completed.



Forward RNN and backward RNN
→ 2 RNNs
RNN ← RNN
Concatenate.

$$h_t^{\rightarrow} = \tanh(\vec{w} h_{t-1}^{\rightarrow} + \vec{u} x_t + \vec{b})$$

$$h_t^{\leftarrow} = \tanh(\overleftarrow{w} h_{t+1}^{\leftarrow} + \overleftarrow{u} x_t + \overleftarrow{b})$$

Forward RNN

Backward RNN

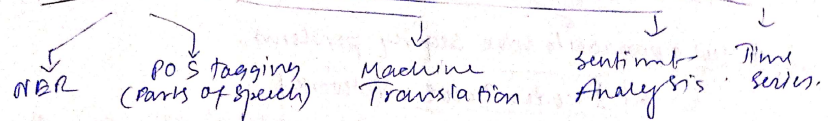
$$\vec{h}_t = \tanh(\vec{w} \vec{h}_{t-1} + \vec{u} x_t + \vec{b})$$

$$\overleftarrow{h}_t = \tanh(\overleftarrow{w} \overleftarrow{h}_{t+1} + \overleftarrow{u} x_t + \overleftarrow{b})$$

$$y_t = \sigma(V[\vec{h}_t, \overleftarrow{h}_t] + b)$$

then concatenating them

Application.



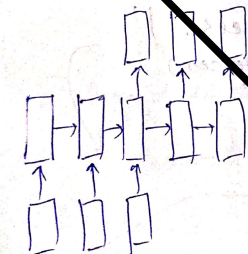
Drawbacks

- Complexity ↑
- Overfitting ↑
- speech recognition . x
- the whole data is not available at once.

Seq2Seq

Epic history of LLMs

Seq2Seq tasks



NLP tasks

- machine translation
- text summarizer
- Question ans. (data with knowledge base)
- chatbot (input text) → output (text)
- speech-to-text

History of Seq2Seq Models

